Network Monitoring System

Points: 35% out of 100%

Machine: 4 virtual machines on either personal computers or public cloud

Group size: 3-4 people

1. Objective

The primary purpose of this work is to give students hands-on experience in system monitoring, which is a common responsibility in system administration.

2. Tasks

Each group needs to install and configure log centralization and network monitoring systems according to the following conditions:

- Both log centralization and network monitoring systems must be open-source software that can be manually installed on Linux.
- Any cloud-based or fully managed services are not allowed.
- For clarity, few examples of network monitoring systems are Zabbix, Zenoss.
- The log centralization and network monitoring systems need to collect information from **at least three Linux virtual machines.**
- Each virtual machine needs to run at least a web server service such as Apache HTTP, NGINX, or OpenLiteSpeed.

Log Centralization System

The log centralization system needs to be able to perform the following:

A) Provide a dashboard that allows its users to query logs.
B) Collect system logs, such as authentication logs and kernel logs.
C) Collect web server access logs.
D) Collect web server error logs.
E) Alerts through e-mails when specified conditions are met.

System states and Metrics

The network monitoring system needs to be able to perform the following:

A) Provide a dashboard that gives an overview of all the system under monitoring.
B) Collect information about and monitor system uptime.
C) Collect information about and monitor service status (e.g. NGINX service status)
D) Collect information about and monitor CPU utilization.

E) Collect information about and monitor memory utilization.
F) Collect information about and monitor network inbound/outbound traffic rate.
G) Collect information about and monitor disk I/O (disk read/write throughput).
H) Collect information about and monitor disk free space.
I) Alerts through e-mails when specified conditions are met.

Alerts

As a minimum, students need to emulate/trigger the specified conditions/scenarios as well as configure log centralization and network monitoring systems to send alerts through e-mails as follows:

A) The number of unauthorized events exceeds a certain threshold, e.g. unauthorized sudo was requested
B) Error or Critical logs appear in web server error logs
C) Web server service fails or the service is stopped
D) CPU utilization exceeds a certain threshold, 75% for example
E) Memory utilization exceeds a certain threshold, 75% for example
F) Network inbound traffic exceeds a certain rate
G) Disk writing throughput exceeds a certain rate/threshold
H) Disk free space is lower than a certain threshold

3. Marking Criteria

No.	Condition(s)	Points
1	Demonstrate access to the configured log centralization system	1
2	Demonstrate access to the configured network monitoring system	1
3	Collect system logs and send alerts based on specified system logs	2
4	Collect web server access/error logs and send alerts based on specified web server access/error logs	2
5	Monitor system uptime and send alerts based on system states	2
6	Monitor web server service and send alerts based on its state	2
7	Monitor CPU utilization and send alerts based on a certain threshold	3
8	Monitor memory utilization and send alerts based on a certain threshold	3
9	Monitor network inbound/outbound traffic rate and send alerts based on a certain rate	3
10	Monitor disk reading/writing throughput and send alerts based on a certain rate	3
11	Monitor disk free space and send alerts based on a certain rate	3
12	Answer instructor’s questions during demonstration	10

4. Submission

Each group of students can make an appointment with an instructor to demonstrate their work in person. Please submit the assignment no later than November 18, 2026.